Nachiappan S K

Phone: (+91) 830-084-3618 Email: nicenach149@gmail.com LinkedIn: Mr-NSKN GitHub: Nachiappan14

Professional Summary

Software engineer with 3+ years at Oracle Cloud Infrastructure, building Tier 0 distributed networking services across backend APIs, infrastructure automation, security, and production operations. Combines computer science fundamentals with hands-on experience taking complex systems from design and implementation through validation, rollout, and incident response. Delivered deployment automation and regional build improvements through cross-team engineering. Brings Python/PyTorch research and applied AI/ML experience in data-drift detection, NLP, and computer vision, alongside AI-assisted development and leadership as an elected Student Affairs Council member at IIIT Bangalore.

Professional Experience

Oracle (OCI / Oracle Cloud Infrastructure) **Aug 2023 – Present**

Core Infrastructure Engineer 2 | Cloud Networking *Bengaluru, India*

- Co-designed Private Access (PA) migration and Private Service Access (PSA) integration for 10+ Cloud@Customer regions, adapting connectivity, certificates, and authorization for low-footprint infrastructure.
- Cut PA build time by 85%, from 1,200 to 180 minutes, by analyzing releases across 3 regions, consolidating deployment stages, and correcting dependency ordering.
- Authored service plans and manifests with 30 PA partner teams and approximately 20 PSA partner teams, resolving bootstrap dependency cycles and coordinating regional deployment requirements.
- Built and rolled out a team-wide autonomous build-test-repair AI agent loop, reducing deployment-change iteration from 2–3 days to 3–4 hours by using historical regional dependencies to update Terraform and deployment plans, orchestrate compilation and lab validation, and iteratively diagnose and repair failures.
- Co-developed PA/PSA APIs for provider-initiated DNS updates, enforcing resource-principal authorization, IAM policies, and endpoint-association checks; strengthened resilience with retries based on response status, exponential backoff, and calibrated throttling.
- Co-engineered IPv6 and dual-stack support across 3 services: Service Gateway, Private Endpoint, and Reverse Connection Endpoint; independently rebuilt canaries with thread pools, stack-specific resources, and Grafana diagnostics. Enabled earlier PSA validation through source-region canaries, co-implemented temporary routing, and implemented subsequent target-region migration and routing cleanup.
- Remediated a Private Endpoint API access-control vulnerability across 85+ production regions, owning investigation, design, implementation, and rollout; identified a separate PA–Work Requests dependency cycle that could deadlock regional recovery and proposed remediation.
- Automated PSA host patching across 110+ production regions by securing onboarding approvals and configuring concurrency limits, minimum serving capacity, and rollout safeguards for availability-aware maintenance.
- Triaged 1,850+ Sev-2 incidents over 3 years across 4 OCI networking control planes, troubleshooting customer-impacting failures and coordinating cross-team mitigation alongside Sev-1 response and service-limit requests.

HP **Feb 2023 – Jul 2023**

Research Intern, Data & Diagnostics - Insights Lab *Bangalore, Karnataka*

- Engineered a Python framework covering 4 drift categories—numerical, categorical, volume, and text—to consolidate detection results into audit data for data-quality reporting.
- Adapted ADWIN with adaptive windows and statistical change verification to detect numerical drift; created volume and categorical detectors using time-based aggregation and category-level analysis.
- Constructed an NLP pipeline combining preprocessing, BERT sentence embeddings, and classification models to identify shifts in textual content over time.

Accenture Labs **May 2022 – Jul 2022**

Research Intern *Bangalore, Karnataka*

- Analyzed software engineering challenges in building and scaling metaverse systems during a research internship at Accenture Labs.

Education

International Institute of Information Technology, Bangalore **Aug 2018 – Jul 2023**

Integrated Master of Technology, Computer Science and Engineering (AI/ML Specialization) *CGPA: 3.41/4*

Additional Experience

Automated Campus Security and Analytics System (sahasraaksha.github.io) | **Jun 2021 – Jul 2023**
Python, PyTorch, React, Node.js, MongoDB

- Designed a CCTV analytics system integrating intrusion detection, attire-based person search, vehicle detection and tracking, and number plate recognition and search, with real-time feed prioritization and alerts through the application and Telegram.
- Piloted the system at NIMHANS, Bengaluru, and demonstrated its surveillance and analytics capabilities at Bangalore Tech Summit 2022.

Technical Skills

Languages: Python, Java, Bash, C, C++, JavaScript.
Web Technologies: React, Node.js, CSS.
Backend & Distributed Systems: REST APIs, workflow orchestration using the OCI workflow engine, error handling, status-aware retries, exponential backoff, API throttling, API authorization, multithreading.
AI/ML: PyTorch, computer vision, deep learning, NLP with BERT embeddings, data-drift detection, LSTM models, unsupervised learning.
AI Development & Automation: Claude Code, OpenAI Codex, autonomous build-test-repair workflows, skill development for on-call assistance.
Cloud & Infrastructure: Oracle Cloud Infrastructure (OCI), Terraform/IaC, Python resource-management tooling, Docker, Linux, Git, multi-region deployment automation.
Reliability & Observability: Production on-call rotations (primary, secondary, SME escalation), incident response and troubleshooting; canary monitoring, logs, metrics, Grafana, alarms, runbooks.
Testing: Maven, JUnit, Mockito, unit testing, integration testing, canary testing.
Security & Networking: mTLS, OCI IAM policies, resource principals, IPv4/IPv6, dual-stack networking, private endpoints, VCNs, subnets.
Data & Databases: SQL, pandas, MongoDB.

Publications

Cost-Efficient SFC Provisioning in Container Environment **Feb 2023**

- Published at the 12th International Conference on Industrial Technology and Management (ICITM), 2023. IEEE publication

Honors & Awards

Magnifying Glass Award | *Oracle Networking* **2026**

- Received Oracle Networking's Magnifying Glass Award for contributions to enhancing the customer experience.

Smart India Hackathon | ***Winner*** **2022**

- Won the problem statement among 70+ applicant teams with an automatic number plate recognition solution using surveillance cameras mounted 3–4 meters above ground.

Cisco thingQbator | *Top 30* **2021**

- Reached the top 30 in the cohort 4 startup track, collaborating in a 4-member team to bring an idea-stage startup to fruition.

Smart India Hackathon | *Finalist* **2020**

- Reached the finals with a solution for a DRDO problem statement, combining unsupervised clustering of airborne objects on synthesized data with LSTM-based trajectory prediction.

Leadership

Student Affairs Council **Sep 2020 – Oct 2021**
Elected Member *IIIT Bangalore*

- Represented students in discussions with university administration as an elected council member; co-organized technical and cultural events at IIIT Bangalore.